

Monetary Economics and Policy

Pierpaolo Benigno

Chapter 1: The Value of Currency



Prepared by Marcos Consuegra

What We Will Learn in This Chapter

- ▶ Understand the distinction between **currency**, **money**, and their core functions in the economy.
- ▶ Study how the **value of currency** is determined in a simple model economy.
- ▶ Explore how **monetary and fiscal policy** interact to anchor the price level.
- ▶ Compare two main frameworks of price determination:
 - ▶ **Fiscal Theory of the Price Level (FTPL)**
 - ▶ **Central Bank Theory of the Price Level (CBTPL)**
- ▶ Assess the role of **real backing** (taxes, assets, gold) in ensuring a stable value of currency.

Outline

- 1 Currency and Money: Concepts
- 2 Model
- 3 Determining the value of the currency
- 4 Fiscal theory of the price level
- 5 Central Bank theory of the price level
- 6 A Model with Gold Holdings
- 7 Takeaways

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Currency: Metallic (Commodity) Regime

- ▶ The term **currency** refers to a well-defined monetary unit issued under a recognized authority.
- ▶ **Example – The Florentine Florin** Struck by the Republic of Florence and circulating between **1252 and 1533**:
 - ▶ A gold coin containing **54 grains** ($\approx 3.499 \text{ g} \approx 0.1125 \text{ troy oz}$) of fine gold.
 - ▶ Featured a consistent alloy and purity.
 - ▶ Bore the emblem of the city (*giglio*) on one side and **Saint John the Baptist** on the reverse.
- ▶ Metallic coins had **intrinsic value**, directly linked to their precious-metal content.

The Florentine Florin



Florentine Florin (Republic of Florence, 13th–16th century).

Obverse: *giglio*; Reverse: Saint John the Baptist.

Currency: Paper-Backed Regime

- ▶ **Example – The U.S. Dollar under the Gold Standard** Formally established by the **Gold Standard Act of 1900** and maintained until 1933.
 - ▶ Each dollar was defined as **23.22 grains of gold** (≈ 1.5046 g).
 - ▶ The dollar was a **liability of the U.S. Treasury** and an **asset for the bearer**.
 - ▶ Gold certificates explicitly stated: *“This certifies that there have been deposited in the Treasury of the United States one dollar in gold coin, payable to the bearer on demand.”*
- ▶ This system linked the value of paper currency to a **fixed quantity of gold**, ensuring convertibility.

U.S. Dollar under the Gold Standard



U.S. Gold Certificate, early 20th century.

Printed statement: "This certifies that there have been deposited in the Treasury of the United States one dollar in gold coin, payable to the bearer on demand."

Intrinsically Worthless Currency

- ▶ The *currency unit* is **not** tied to any commodity.
- ▶ An **intrinsically worthless currency** is a claim that makes no other promise than to be redeemable in units of itself.
- ▶ A dollar is simply a claim to ...**another dollar!**
- ▶ “\$1” represents one unit of *central bank (CB) liabilities*.
- ▶ The CB can create such liabilities **at will**, subject only to its policy objectives.

U.S. Dollar Today



British Pound Today



Printed statement: "Bank of England. I promise to pay the bearer on demand the sum of five pounds."

Three Properties of Currency

- ① **Unit of account** — the numéraire for prices and contracts.
- ② **Medium of exchange** — the means of payment in transactions.
- ③ **Store of value** — maintains its value across time and contingencies (i.e., keeps its promises).

Remarks.

- ▶ The *unit of account* is not an essential property of currency.
- ▶ In medieval times, the unit of account was often an *imaginary unit* that did not circulate physically.
- ▶ The *store-of-value* property should not be confused with purchasing power:
 - ▶ A gold coin guaranteed its **intrinsic metal content**, not its purchasing power.
 - ▶ A modern dollar banknote maintains its **nominal dollar value** over time and is a perfect store of value.

Money

- ▶ **Money** consists of claims on currency that share its key properties — being a **medium of exchange** and a **store of value** — to varying degrees.
- ▶ **Currency is money, but money is not necessarily currency.**
- ▶ A deposit at a commercial bank is not currency, but it is a **claim on currency**. It functions as money because it is used as a medium of exchange and serves as a store of value — as long as the issuer (the bank) can honor redemption. For the issuer, however, redeeming deposits is not as trivial as it is for the central bank.

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Model Overview

Economic Setting

- ▶ Infinite-horizon, closed economy with perfect foresight
- ▶ Single consumption good C with constant endowment Y

Agents

- ▶ Representative consumer
- ▶ Treasury
- ▶ Central Bank (CB)

One Currency = CB's liabilities

- ▶ Cash M_t
- ▶ Reserves X_t

Representative Consumer: Preferences

Intertemporal Utility

$$\sum_{t=t_0}^{\infty} \beta^{t-t_0} U(C_t) \quad (1.1)$$

where β with $0 < \beta < 1$ is the subjective discount factor in preferences.

Properties of $U(C_t)$

- ▶ $U'(C_t) > 0$
- ▶ $U''(C_t) < 0$

Budget Constraint

Budget Constraint at time t

$$\frac{B_t}{1+i_t} + \frac{X_t}{1+i_t^X} + M_t + P_t C_t + T_t = B_{t-1} + X_{t-1} + M_{t-1} + P_t Y \quad (1.2)$$

Where

- ▶ B_t : default-free one-period bond (unit face value), discounted at nominal rate i_t
- ▶ X_t : CB reserves, paying i_t^X
- ▶ M_t : cash (zero interest)
- ▶ P_t : price of consumption good
- ▶ T_t : lump-sum taxes (negative = transfers)
- ▶ Y : constant endowment of goods

Portfolio constraints

- ▶ Bonds: B_t can be held as asset (> 0) or issued as debt (< 0)
- ▶ Cash and reserves: $M_t \geq 0, X_t \geq 0 \forall t \geq t_0$

Transmission Mechanism of the Policy Rate

- ▶ The central bank (CB) sets the interest rate on reserves, denoted by i_t^X .
- ▶ The CB supplies a positive amount of reserves: $X_t > 0$.
- ▶ Bonds B_t and reserves X_t are both default-free and have identical payoffs.

No-Arbitrage Condition

Since reserves are in **positive supply**, no-arbitrage implies that the **market** interest rate on default-free securities must equal the interest rate on reserves:

$$i_t = i_t^X.$$

Key implication: By choosing i_t^X —the interest rate on its liabilities, which are default-free by definition of currency—the CB effectively determines the short-term interest rate on any other default-free security.

- ▶ From now on, i_t denotes the policy rate.

Cashless Equilibrium and the Zero Lower Bound

- ▶ Cash and bonds are **perfect substitutes** as media of exchange (see budget constraint (1.2)).
- ▶ Bonds pay a non-negative interest rate, while cash yields zero.
- ▶ $\Rightarrow$ When $i_t > 0$, bonds dominate cash, and money demand is zero.
- ▶ The existence of cash implies the **zero lower bound (ZLB)**:

$$i_t \geq 0.$$

- ▶ *Proof by contradiction:* If $i_t < 0$, consumers could borrow ($B_t < 0$), invest in cash, pay back debt, and earn a profit with certainty.

Cashless Equilibrium and the Zero Lower Bound

Results

- 1 The option to hold cash sets a zero lower bound on the nominal interest rate:
 $i_t \geq 0$.
- 2 In equilibrium:

$$M_t = 0 \quad \forall t \geq t_0,$$

unless $i_t = 0$.

- From now on, we assume a **cashless equilibrium**:

$$M_t = 0 \quad \text{for all } t \geq t_0.$$

Borrowing Limit

- ▶ Consumers cannot borrow indefinitely (i.e. hold a negative B_{t-1} forever).
- ▶ Moreover, their debt—carrying a default-free interest rate i_t —must be repaid with certainty.

Nominal borrowing limit:

$$\underbrace{-B_{t-1}}_{\text{debt}} \leq \underbrace{X_{t-1}}_{\text{assets}} + \underbrace{\sum_{j=0}^{\infty} \tilde{R}_{t,t+j} (P_{t+j} Y_{t+j} - T_{t+j})}_{\text{present discounted value of net income}} < \infty, \quad (1.3)$$

where $\tilde{R}_{t,t+j}$ is the nominal discount factor from period $t+j$ with respect to time t :

$$\tilde{R}_{t,t+j} = \prod_{i=1}^j \frac{1}{1+i_{t+i-1}}, \quad \tilde{R}_{t,t} \equiv 1, \quad j \geq 1. \quad (1.4)$$

- ▶ In equation (1.3), debt cannot exceed available assets and the PDV of net income.

Borrowing Limit in Real Terms

- The nominal borrowing limit in (1.3) can be expressed in real terms:

Real borrowing limit:

$$-\frac{B_{t-1}}{P_t} \leq \frac{X_{t-1}}{P_t} + \sum_{j=0}^{\infty} R_{t,t+j} \left(Y - \frac{T_{t+j}}{P_{t+j}} \right) < \infty, \quad (1.5)$$

where $R_{t,t+j}$ is the real discount factor from $t+j$ w.r.t. time t .

Real discount factor:

$$R_{t,t+j} = \prod_{i=1}^j \frac{1}{1 + r_{t+i-1}}, \quad R_{t,t} \equiv 1, \quad j \geq 1. \quad (1.6)$$

- Note, that the real interest rate r is defined as $1 + r_t \equiv (1 + i_t) \frac{P_t}{P_{t+1}}$.

Intertemporal Budget Constraint

- ▶ The borrowing limit in (1.5) is **equivalent** to the household's intertemporal budget constraint, using eq. (1.2).

Intertemporal Budget Constraint:

$$\sum_{t=t_0}^{\infty} R_{t_0,t} C_t \leq \frac{B_{t_0-1} + X_{t_0-1}}{P_{t_0}} + \sum_{t=t_0}^{\infty} R_{t_0,t} \left(Y - \frac{T_t}{P_t} \right). \quad (1.7)$$

- ▶ The present discounted value (PDV) of consumption cannot exceed:
 - ▶ the initial real value of financial assets, and
 - ▶ the PDV of real net income.

See Proof in Appendix

Asset Limit Condition

- Constraints (1.5) or (1.49) can be equivalently expressed, using eq. (1.2), as an asset limit constraint.

Asset Limit Constraint:

$$\lim_{t \rightarrow \infty} \left\{ R_{t_0, t} \left(\frac{B_{t-1} + X_{t-1}}{P_t} \right) \right\} \geq 0. \quad (1.8)$$

- This condition requires that the discounted value of the household's asset holdings remain non-negative in the limit.

Proof available in Appendix

Consumer Optimality: Euler Equation

- The consumer's intertemporal optimization problem yields the **Euler equation**.

Euler Equation:

$$\underbrace{U_c(C_t)}_{\text{marginal utility of consumption}} = \underbrace{\beta (1 + r_t) U_c(C_{t+1})}_{\text{marginal utility of saving}}, \quad (1.9)$$

where $U_c(C_t)$ denotes the marginal utility of consumption at time t .

- At the optimum, the marginal utility of consuming one unit today equals the marginal utility of saving that unit for next period.

See Proof in Appendix

Consumer Optimality: IBC or Transversality Condition

- At the optimum, consumers exhaust all resources $\rightarrow$ the intertemporal budget constraint holds with equality.

Intertemporal Budget Constraint:

$$\sum_{t=t_0}^{\infty} R_{t_0,t} C_t = \frac{B_{t_0-1} + X_{t_0-1}}{P_{t_0}} + \sum_{t=t_0}^{\infty} R_{t_0,t} \left(Y_t - \frac{T_t}{P_t} \right), \quad (1.10)$$

- Or, equivalently, the transversality condition holds, i.e. eq. ((1.8)) with equality:

Transversality Condition:

$$\lim_{t \rightarrow \infty} R_{t_0,t} \frac{B_{t-1} + X_{t-1}}{P_t} = 0. \quad (1.11)$$

Central Bank

- ▶ In this cashless economy, the CB:
 - ▶ **issues CB reserves** X^C (the only liability)
 - ▶ **holds assets** B^C , private or treasury bonds
 - ▶ **makes transfers** T^C to the treasury (positive = remittances, negative = subsidies)

CB Flow Budget Constraint:

$$\frac{B_t^C}{1+i_t} - \frac{X_t^C}{1+i_t} = B_{t-1}^C - X_{t-1}^C - T_t^C. \quad (1.12)$$

Central Bank: Key Properties

- ▶ Reserves X^C are claims on themselves; they can **always** be paid, even if $B_t^C = 0 \longrightarrow$ **No** solvency condition is required on X^C .
- ▶ Given X_{t-1}^C , the CB can freely choose both the quantity of reserves X_t^C and the interest rate i_t .
- ▶ By choosing the transfers policy T_t^C , asset holdings B_t^C adjust according to the flow budget constraint (1.12).

Treasury Budget Constraint

- ▶ The Treasury issues debt. At time t must pay its outstanding obligations B_{t-1}^F by:
 - ▶ Lump-sum taxes T_t (net of transfers)
 - ▶ CB remittances T_t^C
 - ▶ Issuance of new debt B_t^F

Treasury Budget Constraint:

$$B_{t-1}^F = T_t + T_t^C + \frac{B_t^F}{1 + i_t} \quad (1.13)$$

- ▶ B_{t-1}^F are claims on currency, **not** currency. They are not, by definition, default-free!
- ▶ To be default free, Treasury debt has to satisfy a solvency constraint.

Treasury Solvency and No-Ponzi Conditions

- Treasury debt should be repaid with certainty to be default free (and paying the default-free rate i):

$$\underbrace{B_{t-1}^F}_{\text{Treasury debt}} \leq \underbrace{\sum_{j=0}^{\infty} \tilde{R}_{t,t+j} (T_{t+j} + T_{t+j}^C)}_{\text{pdv of the Treasury's resources}} < \infty$$

In real terms:

$$\frac{B_{t-1}^F}{P_t} \leq \sum_{j=0}^{\infty} R_{t,t+j} \left(\frac{T_{t+j}}{P_{t+j}} + \frac{T_{t+j}^C}{P_{t+j}} \right) < \infty \quad (1.14)$$

or equivalently **Limit Condition:**

$$\lim_{t \rightarrow \infty} R_{t_0,t} \frac{B_{t-1}^F}{P_t} \leq 0 \quad (1.15)$$

Treasury - Implications for the Model

- ▶ The treasury cannot accumulate debt indefinitely; it must be covered by taxes and CB remittances in present value.
- ▶ If taxes are reduced in certain periods, they must increase in later periods to maintain solvency.
- ▶ For the subsequent analysis, we assume that condition (1.14) or, equivalently, (1.15) holds with equality.
- ▶ The equality assumption is supported by political-economic considerations (excessive taxation is costly) or by the rationale that the treasury does not intend to accumulate a positive long-term asset position.

Market Clearing Conditions

- ▶ Debt issued by the government (B^F) is held either by the CB (B^C) or the consumer (B). CB reserves (X^C) are held only by the consumer (X).

Asset Market Clearing:

$$B_t^F = B_t^C + B_t \quad (1.16)$$

$$X_t^C = X_t \quad (1.17)$$

Goods Market Clearing:

$$C_t = Y$$

- ▶ Consumption equals output, consistent with a closed economy with no investment and no government purchases.

See Proof in Appendix

Equilibrium Condition: Fisher Equation

- ▶ Substituting $C_t = Y$ into the Euler equation (1.9) delivers the Fisher equation:

Fisher Equation:

$$\underbrace{1 + i_t}_{\text{nominal rate}} = \underbrace{\frac{1}{\beta}}_{\text{real rate}} \underbrace{\frac{P_{t+1}}{P_t}}_{\text{inflation}}, \quad (1.18)$$

- ▶ This condition links the nominal interest rate to the real rate and to next period inflation.
- ▶ Note: with a constant endowment, the real interest rate is constant and equal to $1/\beta$.

Equilibrium Condition: Intertemporal Resource Constraint

- ▶ Using $C_t = Y$ and the Fisher equation (1.18), the intertemporal budget constraint (1.10) can be written as the:

Intertemporal Resource Constraint

$$\frac{B_{t_0-1} + X_{t_0-1}}{P_{t_0}} = \sum_{t=t_0}^{\infty} \beta^{t-t_0} \frac{T_t}{P_t} \quad (1.19)$$

- ▶ **LHS:** Real net government liabilities to the private sector at t_0 (including both Treasury debt and CB liabilities).
- ▶ **RHS:** Present-discounted value of real taxes collected.
- ▶ The intertemporal resource constraint is not a solvency constraint but an **equilibrium condition**.

Equilibrium Condition: Treasury Solvency

Treasury's solvency condition is given by (1.14), stated with equality and using the Fisher equation (1.18):

Treasury Solvency Condition:

$$\frac{B_{t_0-1}^F}{P_{t_0}} = \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left(\frac{T_t}{P_t} + \frac{T_t^C}{P_t} \right) \quad (1.20)$$

Treasury solvency requires that the stock of outstanding real debt be backed by the present discounted value of real taxes at any equilibrium path of prices and CB remittances.

Equilibrium Conditions: Budget Constraints

CB Budget Constraint in Equilibrium:

$$\frac{X_t - B_t^C}{1 + i_t} = X_{t-1} - B_{t-1}^C + T_t^C \quad (1.21)$$

Treasury Budget Constraint in Equilibrium:

$$\frac{B_t^F}{1 + i_t} = B_{t-1}^F - T_t - T_t^C \quad (1.22)$$

Equilibrium Definition

Equilibrium: A set of sequences

$$\{P_t, i_t, X_t, B_t^C, T_t^C, B_t^F, T_t, B_t\}_{t=t_0}^{\infty}$$

with $\{P_t, i_t, X_t, B_t^C\}_{t=t_0}^{\infty} \geq 0$ that satisfies:

- ▶ Bond market clearing (1.16)
- ▶ Euler equation (1.18)
- ▶ Intertemporal resource constraint (1.19)
- ▶ Government budget constraints (1.21), (1.22)

given initial conditions $B_{t_0-1}^C, B_{t_0-1}^F, X_{t_0-1}$. Taxes $\{T_t\}_{t=t_0}^{\infty}$ must satisfy the solvency condition (1.20) for any equilibrium sequences $\{P_t, T_t^C\}_{t=t_0}^{\infty}$, given the initial condition $B_{t_0-1}^F$.

Government Policy

- ▶ Equilibrium characterization shows 4 degrees of freedom, which are the policy choices that the government can take.
- ▶ CB chooses sequences $\{i_t, X_t, T_t^C\}_{t=t_0}^{\infty}$, with asset holdings adjusting via (1.21).
- ▶ Treasury sets taxes $\{T_t\}_{t=t_0}^{\infty}$ to satisfy (1.20) at *any* equilibrium sequences for prices $\{P_t\}_{t=t_0}^{\infty}$ and CB remittances $\{T_t^C\}_{t=t_0}^{\infty}$.

Monetary Policy Insights

Key Insights:

- ▶ The economy does not have an inherent tendency to select a unique price level without the CB specifying policy sequences $\{i_t, X_t, T_t^C\}_{t=t_0}^{\infty}$.
- ▶ Forward guidance is important: policymakers must communicate both current and future paths of policy to eventually be able to determine equilibrium.
- ▶ Balance-sheet policies matter: specifying interest rates alone is not sufficient to uniquely determine the price level (as shown next).

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Value of the Currency: Limits of Interest-Rate Policy

- ▶ Interest-rate policy alone cannot determine the currency's value.
- ▶ Using the Fisher equation (1.18), if the CB pegs $i_t = i$, only the rate of price changes is determined, i.e. $\frac{P_{t+1}}{P_t}$, not the price level itself.
- ▶ This indeterminacy result also holds with active policies reacting to deviations from a target $\bar{P}$ as the following:

Policy Rule

$$1 + i_t = \max \left\{ \frac{1}{\beta} \left(\frac{P_t}{\bar{P}} \right)^\phi, 1 \right\}, \quad \phi \geq 0 \quad (1.23)$$

- ▶ In (1.23), the interest rate is equal to the real rate $1/\beta$ and reacts with a coefficient ϕ to the ratio of the current price level with respect to the target $\bar{P}$, unless the zero lower bound is achieved.

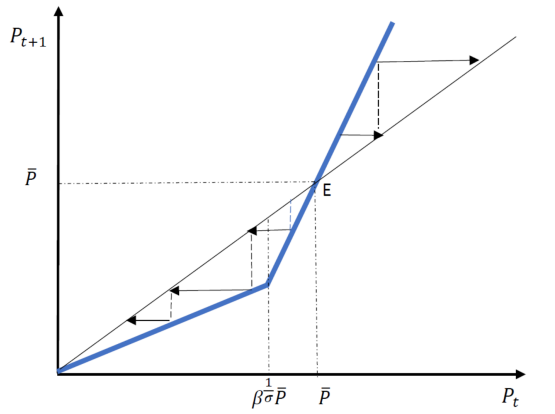
Price-Level Indeterminacy

- ▶ Combining the Fisher equation (1.18) and the interest-rate rule (1.23) gives a non-linear difference equation for the price level:

$$\frac{P_{t+1}}{P_t} = \max \left\{ \left(\frac{P_t}{\bar{P}} \right)^\phi, \beta \right\} \quad (1.24)$$

- ▶ For $\phi > 0$, infinite solutions exist, all indexed by $P_{t_0} \in (0, \infty)$:
 - ▶ Stationary solution exists only if $P_{t_0} = \bar{P}$
 - ▶ $P_{t_0} > \bar{P} \Rightarrow$ inflationary path; $P_{t_0} < \bar{P} \Rightarrow$ deflationary path
- ▶ For $\phi = 0$, infinite stationary solutions exist, all indexed by $P_{t_0} \in (0, \infty)$
- ▶ **Conclusion:** Interest-rate rules, even Taylor-type rules, cannot uniquely determine the price level without specifying appropriately additional policy instruments (*the other degrees of freedom*).

Phase Diagram Illustration



Phase diagram for $\phi > 0$. The stationary solution is $P_t = \bar{P}$.

If $P_t \leq \beta^{1/\phi} \bar{P}$, the deflation rate is β (with $0 < \beta < 1$).

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Fiscal Theory of the Price Level (FTPL) – Concept

- ▶ FTPL critically relies on the **coordination** between monetary and fiscal policy.
- ▶ **Distinction between Treasury and Central Bank (CB) liabilities:**
 - ▶ **Treasury debt** promises payment in a currency for which the Treasury has **limited resources**.
 - ▶ **Central bank liabilities** are redeemable in **units of themselves**, and can therefore be issued in **unlimited supply**.
- ▶ In FTPL the CB **fully guarantees** treasury debt. Distinction between treasury and CB is now blurred.
- ▶ Solvency condition (1.14) no longer constraints the path of taxes.
- ▶ The government budget constraints are now consolidated, by using (1.12), (1.13) and bond market clearing (1.16):

Consolidated Government Budget Constraint

$$\frac{B_t + X_t}{1 + i_t} = B_{t-1} + X_{t-1} - T_t \quad (1.25)$$

Equilibrium under FTPL

Given that tax policy is no longer constrained by solvency condition (1.20), equilibrium is characterized by a set of sequences:

$$\{i_t, P_t, X_t, T_t, B_t\}_{t=t_0}^{\infty}$$

with $\{P_t, i_t, X_t\}_{t=t_0}^{\infty} \geq 0$ that satisfies:

- ▶ Euler Equation (1.18)
- ▶ Intertemporal resource constraint (1.19)
- ▶ Interest-rate rule (1.23)
- ▶ Consolidated government budget constraint (1.25)

given initial conditions X_{t_0-1}, B_{t_0-1} .

Two policy instruments must be specified in addition to the interest rate:

- ▶ The sequence of taxes $\{T_t\}_{t=t_0}^{\infty}$
- ▶ The sequence of reserves $\{X_t\}_{t=t_0}^{\infty}$, taken as arbitrary positive.

Constant Real Tax Policy

- ▶ The key factor in influencing the price level is the formulation of the tax policy.
- ▶ Consider the constant real tax policy:

$$\frac{T_t}{P_t} = (1 - \beta)\tau, \quad \tau > 0 \quad (1.26)$$

- ▶ Substituting into the intertemporal budget condition (1.19) gives:

$$P_{t_0} = \frac{B_{t_0-1} + X_{t_0-1}}{\tau} \quad (1.27)$$

Price Level Determination

- ▶ Equation (1.27) pins down the initial price level which implies a proportional relationship between government net liabilities and the price level.
- ▶ The entire path $\{P_t\}_{t>t_0}^{\infty}$ is then uniquely determined by (1.24), given P_{t_0} .
- ▶ This type of policy is called **active** because taxes are fixed in real terms and do not adjust to ensure solvency. Indeed, solvency is **guaranteed** by the central bank.

Tax Policy and Equilibrium Selection

- ▶ Among multiple possible paths from (1.24) given (1.23), tax policy can select the equilibrium one.
- ▶ Define

$$\tau^* = \frac{B_{t_0-1} + X_{t_0-1}}{\bar{P}}$$

- ▶ Cases:
 - ▶ $\tau = \tau^* \Rightarrow P_{t_0} = \bar{P}$ (stationary equilibrium)
 - ▶ $\tau > \tau^* \Rightarrow P_{t_0} < \bar{P}$ (deflationary path if $\phi > 0$)
 - ▶ $\tau < \tau^* \Rightarrow P_{t_0} > \bar{P}$ (inflationary path)
- ▶ **Conclusion:** Under FTPL, tax policy can uniquely anchor the price level.

Role of the Central Bank in the FTPL

- ▶ **Central Bank (CB) backing:** Solvency is no longer a constraint for the Treasury once the CB guarantees its debt.
- ▶ **Example:** Substituting the tax policy (1.26) into the solvency condition (1.14), and assuming $T_t^C = 0$ for all t , solvency requires:

$$\frac{B_{t_0-1}^F}{P_{t_0}} \leq \tau. \quad (1.28)$$

- ▶ If the equilibrium price level P_{t_0} is too low relative to debt obligations (i.e. the LHS of (1.28) $> \tau$), the Treasury is **insolvent** without the backing of the CB.
 - ▶ To restore solvency, the Treasury would need to raise taxes (τ), and change its tax policy.
- ▶ **Implication:** Fiscal policy can determine the price level only if the CB fully backs the Treasury's debt.
- ▶ **Open question:** Why should the CB back the Treasury and allow it to influence the price level?

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Central Bank Theory of the Price Level (CBTPL) – Concept

- ▶ Unlike the FTPL, the CBTPL relies on the central bank (CB) to **independently determine the price level**.
- ▶ **Key practical cases:**
 - ▶ Private currencies with no connection to the government.
 - ▶ The European Monetary Union (EMU): Treasuries and the CB are separate; solvency constraints for Treasuries are binding (at least before the 2011 sovereign debt crisis).
- ▶ Without CB backing, the Treasury must satisfy the solvency condition (1.14). Assuming equality in (1.14) and substituting into the intertemporal resource constraint (1.19) yields:

$$\sum_{t=t_0}^{\infty} \beta^{t-t_0} \frac{T_t^C}{P_t} = \frac{B_{t_0-1}^C - X_{t_0-1}}{P_{t_0}}. \quad (1.29)$$

Equilibrium under CBTPL

Equilibrium under CBTPL: An equilibrium is a set of sequences:

$$\{i_t, P_t, X_t, B_t^C, T_t^C\}_{t=t_0}^{\infty}$$

with $\{P_t, i_t, X_t\}_{t=t_0}^{\infty} \geq 0$, satisfying:

- ▶ Euler equation (1.18)
- ▶ Interest-rate policy (1.23)
- ▶ CB flow budget constraint (1.12)
- ▶ Intertemporal resource constraint (1.29)

given initial conditions $B_{t_0-1}^C, X_{t_0-1}$.

Two policy instruments must be specified by the CB in addition to the interest rate:

- ▶ Sequence of reserves $\{X_t\}_{t=t_0}^{\infty}$, taken as arbitrary positive
- ▶ Sequence of non-negative remittances $\{T_t^C\}_{t=t_0}^{\infty}$ (to prevent treasury support)

Discussion of the Intertemporal Resource Constraint

- ▶ The equilibrium condition (1.29) deserves further discussion.
- ▶ The present discounted value of CB remittances in *equilibrium* equals the real net assets of the CB.
- ▶ This is **not** a solvency condition, since nominal liabilities are always paid.
- ▶ It is an equilibrium (intertemporal) resource constraint.
- ▶ The CB can determine the initial price level with an appropriate transfer policy.
- ▶ The intertemporal resource constraint holds because prices adjust to the chosen remittance policy, not because the CB actively adjusts remittances.

Real Transfer Policy

- ▶ Consider a real transfer policy (i.e. a commitment to a sequence of remittances indexed to the price level) : $T_t^C/P_t = (1 - \beta)\tau^C$, $\tau^C > 0$.
- ▶ Substituting into (1.29) determines the initial price level:

$$P_{t_0} = \frac{B_{t_0-1}^C - X_{t_0-1}}{\tau^C} \quad (1.30)$$

- ▶ Price is positive if $B_{t_0-1}^C > X_{t_0-1}$, i.e., CB starts with positive net worth.
- ▶ Setting $\tau^C = (B_{t_0-1}^C - X_{t_0-1})/\bar{P}$ pins $P_{t_0} = \bar{P}$, independent of ϕ in (1.23).
- ▶ Unlike FTPL, determination of the price level does not require cooperation between fiscal and monetary authorities.

CBTPL: Net Worth and Key Equation

- ▶ The equilibrium condition (1.29) helps illustrate the intuition behind **price determination**.

- ▶ **Define CB net worth:**

$$N_t^C = \frac{B_t^C - X_t}{1 + i_t}. \quad (1.31)$$

- ▶ Rewriting (1.29) yields:

$$\underbrace{\frac{1}{P_{t_0}}(1 + i_{t_0-1}) N_{t_0-1}^C}_{\text{real value of CB equity}} = \underbrace{\sum_{t=t_0}^{\infty} \beta^{t-t_0} \frac{T_t^C}{P_t}}_{\text{PDV of real remittances}}. \quad (1.32)$$

- ▶ The term $\frac{1}{P_{t_0}}$ represents the **price of CB equity** in units of goods.

CBTPL: Interpretation and Policy Requirements

- ▶ Equation (1.32) highlights the link between the CB's balance sheet and the determination of the price level.
- ▶ Two key requirements for **price-level determination**:
 - ▶ The CB must receive an initial **transfer** at time t_0 (from the Treasury or the private sector) to ensure positive net worth.
 - ▶ The CB must hold **default-free securities** (e.g., B^C) to back its liabilities.
- ▶ Intuition: The value of money—hence the price level—is tied to the **real backing** of the CB's equity and the expected flow of future remittances.

Requirement 1 – Initial Transfer

- ▶ The CB cannot build net worth on its own → it must receive an initial transfer to ensure an initial positive net worth.
- ▶ Rewriting the CB's budget constraint (1.12) using the definition of net worth, N_t^C , via eq. (1.31) :

$$N_t^C = N_{t-1}^C + \Psi_t^C - T_t^C, \quad \Psi_t^C = i_{t-1} N_{t-1}^C$$

- ▶ If $N_{t-1}^C = 0$, then profits $\Psi_t^C = 0$ for all t and N_t cannot be positive.
- ▶ An initial transfer (real or nominal) from the Treasury or private sector is required to give the CB positive equity.
- ▶ Remark: This requirement can be relaxed in models where the CB holds gold or collects seigniorage.

Requirement 2 – Default-Free Securities

- ▶ Equation (1.30) shows that **price determination depends on the presence of default-free assets** in the CB's portfolio.
- ▶ These assets must **exceed the value of liabilities** to ensure positive net worth and to generate profits.
- ▶ Holding **default-free nominal securities** is essential for maintaining price-level control.

Implications

- ▶ A **paper currency** requires a form of **real backing**.
- ▶ With **governmental coordination**, such backing can be achieved through the power of **real taxes** imposed by the Treasury.
- ▶ Without coordination, the **central bank (CB)** can provide real backing through a policy of distributing real earnings, which requires:
 - ▶ an **initial transfer** to ensure positive net worth, and
 - ▶ holding **default-free securities** to generate profits.
- ▶ The **CBTPL** therefore shows how the CB alone can **anchor the price level**, provided it begins with **positive net worth**.

Example: Nominal Remittances Policy

- ▶ Considering a nominal remittances policy:

$$T_t^C = (1 - \beta) (B_{t_0-1}^C - X_{t_0-1}^C), \quad B_{t_0-1}^C > X_{t_0-1}^C$$

- ▶ Substituting again into (1.29) yields:

$$(1 - \beta) \sum_{t=t_0}^{\infty} \beta^{t-t_0} \frac{1}{P_t} = \frac{1}{P_{t_0}} \quad (1.33)$$

- ▶ Considering interest-rate policy (1.23):
 - ▶ **Case 1:** Pure pegging ($\phi = 0$) $\Rightarrow$ no determination of P_{t_0} .
 - ▶ **Case 2:** Active policy ($\phi > 0$) $\Rightarrow$ two equilibria: 1) $P_t = \bar{P}$ and 2) $P_t = +\infty$.

Implications

- ▶ With a nominal remittances policy:
 - ▶ An active interest-rate policy may be necessary to pin down the price level.
 - ▶ A non-monetary equilibrium exists in (1.33): $P_t = +\infty$, where the currency is worthless.
- ▶ This contrasts with a real remittances policy, which uniquely determines the initial price level via CB net worth.
- ▶ Nominal transfer policies alone do not guarantee a unique price level (same under the FTPL).
- ▶ However, allowing the CB to trade in gold can remove the non-monetary equilibrium and uniquely determine prices without relying on Treasury transfers.

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A Model with Gold Holdings

- ▶ Extend the previous framework to include a durable commodity, **gold**.
- ▶ Consumers derive utility from gold, which also forms part of the CB's assets.
- ▶ Gold management allows the CB to control the price level even:
 - ▶ under a nominal remittances policy, and
 - ▶ with periods of negative net worth.
- ▶ Full convertibility into gold is **not required**, unlike a Gold Standard (see Chapter 2).

Representative Consumer: Preferences with Gold

Intertemporal Utility

$$\sum_{t=t_0}^{\infty} \beta^{t-t_0} [U(C_t) + Z(g_t)] \quad (1.34)$$

where g_t represents the households gold holdings providing utility.

Properties

- ▶ $U'(C_t) > 0, U''(C_t) < 0$
- ▶ $Z'(g_t) > 0, Z''(g_t) < 0$

Household Budget Constraint with Gold

Budget Constraint at time t

$$\frac{B_t + X_t}{1 + i_t} + P_{g,t}g_t + P_t C_t + T_t = B_{t-1} + X_{t-1} + P_{g,t}g_{t-1} + P_t Y + P_{g,t}(g_t^S - g_{t-1}^S) \quad (1.35)$$

Where

- ▶ $P_{g,t}$: price of gold in units of currency
- ▶ g_t^S : total gold in the economy owned by the households
- ▶ Other variables are defined as in Budget Constraint

Optimal Demand of Gold

- Optimality condition with respect to gold holdings implies that:

$$\frac{P_{g,t}}{P_t} = \frac{Z_g(g_t)}{U_c(C_t)} + \frac{1}{1+r_t} \frac{P_{g,t+1}}{P_{t+1}} \quad (1.36)$$

where $Z_g(\cdot)$ is the derivative of $Z(\cdot)$ with respect to gold holdings g_t .

- The real price of gold equals the marginal rate of substitution between gold and consumption plus the discounted future real price.

Intertemporal Budget Constraint with Gold

Intertemporal Budget Constraint with Gold

$$\sum_{t=t_0}^{\infty} R_{t_0,t} C_t = \frac{\mathcal{W}_{t_0}}{P_{t_0}} + \sum_{t=t_0}^{\infty} R_{t_0,t} \left\{ \left(Y - \frac{T_t}{P_t} \right) + \frac{Z_g(g_t)}{U_c(C_t)} (g_t^S - g_t) \right\} \quad (1.37)$$

Initial Wealth

$$\mathcal{W}_{t_0} \equiv B_{t_0-1} + X_{t_0-1} + P_{g,t_0} (g_{t_0-1} - g_{t_0-1}^S) \quad (1.38)$$

- The last term on the RHS of (1.37) represents the gains that the household gets by selling its gold holdings.

Additional Equilibrium Conditions

- ▶ CB budget constraint with gold holdings:

$$\frac{B_t^C - X_t}{1 + i_t} + P_{g,t} g_t^C = B_{t-1}^C - X_{t-1} + P_{g,t} g_{t-1}^C - T_t^C. \quad (1.39)$$

- ▶ For simplicity, assume a constant gold endowment, g^S .
- ▶ Gold market equilibrium condition:

$$g^S = g_t + g_t^C, \quad (1.40)$$

where g_t^C are the CB's gold holdings and g_t are households' gold holdings.

CBTPL in an Economy with Gold

- ▶ Consider the case in which the CB does not back the treasury.
- ▶ Starting from the interest-rate rule (1.23) and equilibrium in goods, asset, and gold markets, together with the treasury solvency condition (1.20) and $R_{t_0,t} = \beta^{t-t_0}$, we obtain using (1.37):

$$\frac{B_{t_0-1}^C + P_{g,t_0} g_{t_0-1}^C - X_{t_0-1}}{P_{t_0}} = \sum_{t=t_0}^{\infty} \beta^{t-t_0} \frac{Z_g(g_t)}{U_c(Y)} g_t^C + \sum_{t=t_0}^{\infty} \beta^{t-t_0} \frac{T_t^C}{P_t} \quad (1.41)$$

- ▶ **LHS:** Real net assets of the CB including gold holdings.
- ▶ **RHS:** Present discounted costs of holding gold and real remittances.
- ▶ Holding gold is costly (lower return than reserves) since it provides utility to households.
- ▶ Note: If $g_t^C = 0 \forall t \geq t_0$, (1.41) reduces to (1.29).

CBTPL in an Economy with Gold

- Define:

$$\mathcal{G}_{t_0-1} = \sum_{t=t_0}^{\infty} \beta^{t-t_0} \frac{Z_g(g^S - g_t^C)}{U_c(Y)} (g_{t_0-1}^C - g_t^C) \quad (1.42)$$

- Using (1.36) and (1.42) in (1.41), we can rewrite the CB intertemporal resource constrain as:

$$\frac{B_{t_0-1}^C - X_{t_0-1}}{P_{t_0}} + \mathcal{G}_{t_0-1} = \sum_{t=t_0}^{\infty} \beta^{t-t_0} \frac{T_t^C}{P_t} \quad (1.43)$$

- Comparison with (1.29):
 - $\mathcal{G}_{t_0-1}$ is independent of P_{t_0} and given $\mathcal{G}_{t_0-1} \neq 0$ prevents the non-monetary equilibrium (infinite price level).
 - A time-varying CB gold holding ensures $\mathcal{G}_{t_0-1} \neq 0$.
 - Constant g^C gives $\mathcal{G}_{t_0-1} = 0 \Rightarrow \text{eq. (1.43)} = \text{eq. (1.29)}$.

CBTPL with Gold: Price Level Implications

- ▶ It follows, that it is possible to determine a unique price level even in the case of a nominal remittances policy and even with interest-rate pegging.
- ▶ For simple policies, e.g., $T_t^C = (1 - \beta)T^C \geq 0$ and $1 + i_t = 1/\beta$, equation (1.43) implies:

$$P_{t_0} = \frac{T^C + (X_{t_0-1} - B_{t_0-1}^C)}{\mathcal{G}_{t_0-1}} \quad (1.44)$$

- ▶ Price level P_{t_0} is positive if numerator and denominator have the same sign.
- ▶ Sufficient condition for $\mathcal{G}_{t_0-1} > 0$: the CB holds an initial stock of gold and sells it over time (or acquires and then sells gold later).

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Main Takeaways

- ▶ The value of currency depends on its **backing** and on the **policies of the issuing authority**.
- ▶ **Interest-rate policy alone** cannot determine the price level — coordination with fiscal policy or balance-sheet policy is essential.
- ▶ Under the **Fiscal Theory of the Price Level**, taxes and government debt jointly determine the value of money.
- ▶ Under the **Central Bank Theory of the Price Level**, the CB can anchor the price level independently — provided it has **positive net worth** and holds **default-free assets**.
- ▶ **Real backing matters**: whether through taxes, CB equity, or gold holdings, a credible anchor is needed for currency stability.

8 Appendix: Derivations

Proof: Borrowing Constraint (1.5)

We show that (1.2) and (1.5) imply (1.49).

Consider budget constraint (1.2) setting $i_t = i_t^X$, $M_t = 0$ and dividing it by P_t

$$\frac{B_t + X_t}{P_t(1 + i_t)} = \frac{B_{t-1} + X_{t-1}}{P_t} + Y - C_t - \frac{T_t}{P_t}.$$

Multiply it by discount factor $R_{t_0,t}$ and sum all the constraints from time t_0 to time k , obtaining

$$R_{t_0,k} \frac{B_k + X_k}{P_k(1 + i_k)} = \frac{B_{t_0-1} + X_{t_0-1}}{P_{t_0}} + \sum_{t=t_0}^k R_{t_0,t} \left(Y_t - C_t - \frac{T_t}{P_t} \right)$$

and therefore

$$R_{t_0,k+1} \frac{B_k + X_k}{P_{k+1}} + \sum_{t=t_0}^k R_{t_0,t} C_t = \frac{B_{t_0-1} + X_{t_0-1}}{P_{t_0}} + \sum_{t=t_0}^k R_{t_0,t} \left(Y_t - \frac{T_t}{P_t} \right). \quad (1.45)$$

Use (1.5) at time k as a substitute for $(B_k + X_k)/P_{k+1}$ on the left-hand side of the above equation to obtain

$$\sum_{t=t_0}^k R_{t_0,t} C_t \leq \frac{B_{t_0-1} + X_{t_0-1}}{P_{t_0}} + \sum_{t=t_0}^{\infty} R_{t_0,t} \left(Y_t - \frac{T_t}{P_t} \right).$$

Note first that the right-hand side is bounded by assumption, see (1.5), and that the left-hand side as a function of k is a monotone non-decreasing sequence since $R_{t_0,t}$ is positive and $C_t \geq 0$. Therefore, this bounded monotone non-decreasing sequence has a limit as $k \rightarrow \infty$, which satisfies the bound. Therefore, equation (1.49) follows:

$$\sum_{t=t_0}^{\infty} R_{t_0,t} C_t \leq \frac{B_{t_0-1} + X_{t_0-1}}{P_{t_0}} + \sum_{t=t_0}^{\infty} R_{t_0,t} \left(Y_t - \frac{T_t}{P_t} \right), \quad (1.46)$$

completing the proof.

[Back to Budget Constraint](#)

Proof: Limit Proof

We show, now, that (1.2) and (1.5) imply (1.8). We have shown that $\sum_{t=t_0}^{\infty} R_{t_0,t} C_t$ is finite and bounded above as in (1.46). We take the limit for $k \rightarrow \infty$ in (1.45) and comparing with (1.46), we obtain (1.8), i.e.

$$\lim_{k \rightarrow \infty} R_{t_0,k+1} \frac{B_k + X_k}{P_{k+1}} \geq 0.$$

The reverse, that (1.2) and (1.8) imply (1.5), is also true. Use (1.2) to obtain (1.45) and consider that

$$\sum_{t=t_0}^k R_{t_0,t} C_t \geq 0,$$

and therefore

$$R_{t_0,k+1} \frac{B_k + X_k}{P_{k+1}} \leq \frac{B_{t_0-1} + X_{t_0-1}}{P_{t_0}} + \sum_{t=t_0}^k R_{t_0,t} \left(Y_t - \frac{T_t}{P_t} \right).$$

Taking the limit on both side of the above inequality for $k \rightarrow \infty$ and using (1.8), the borrowing limit (1.5) follows.

[Back to Limit Constraint](#)

Proof: Euler Equation

A representative household maximizes lifetime utility:

$$\max_{\{C_t, B_t, X_t\}_{t=t_0}^{\infty}} \sum_{t=t_0}^{\infty} \beta^{t-t_0} U(C_t) \quad (1.47)$$

subject to the per-period budget constraint

$$\frac{B_t}{1+i_t} + \frac{X_t}{1+i_t} + P_t C_t + T_t = B_{t-1} + X_{t-1} + P_t Y, \quad (1.48)$$

and the (natural) borrowing limit

$$-\frac{B_{t-1}}{P_t} \leq \frac{X_{t-1}}{P_t} + \sum_{j=0}^{\infty} R_{t,t+j} \left(Y - \frac{T_{t+j}}{P_{t+j}} \right), \quad (1.49)$$

where $R_{t,t+j}$ is the real discount factor (with $R_{t,t} = 1$).

Lagrangian

Let λ_t be the multiplier on (1.48) and $\tilde{\zeta}_t \geq 0$ the multiplier on (1.49). Writing the borrowing limit as $0 \leq B_{t-1} + X_{t-1} + P_t \sum_{j \geq 0} R_{t,t+j} \left(Y - \frac{T_{t+j}}{P_{t+j}} \right)$, the Lagrangian is

$$\begin{aligned} \mathcal{L} = \sum_{t=t_0}^{\infty} \beta^{t-t_0} & \left[U(C_t) + \lambda_t \left(B_{t-1} + X_{t-1} + P_t Y - \frac{B_t}{1+i_t} - \frac{X_t}{1+i_t} - P_t C_t - T_t \right) \right. \\ & \left. + \tilde{\zeta}_t \left(B_{t-1} + X_{t-1} + P_t \sum_{j=0}^{\infty} R_{t,t+j} \left(Y - \frac{T_{t+j}}{P_{t+j}} \right) \right) \right]. \end{aligned} \quad (1.50)$$

First-Order Conditions

$$\frac{\partial \mathcal{L}}{\partial C_t} : \quad \beta^{t-t_0}(U_c(C_t) - \lambda_t P_t) = 0 \Rightarrow \lambda_t = \frac{U_c(C_t)}{P_t}, \quad (1.51)$$

$$\frac{\partial \mathcal{L}}{\partial B_t} : \quad -\beta^{t-t_0} \frac{\lambda_t}{1+i_t} + \beta^{t+1-t_0}(\lambda_{t+1} + \tilde{\zeta}_{t+1}) = 0, \quad (1.52)$$

$$\frac{\partial \mathcal{L}}{\partial X_t} : \quad -\beta^{t-t_0} \frac{\lambda_t}{1+i_t} + \beta^{t+1-t_0}(\lambda_{t+1} + \tilde{\zeta}_{t+1}) = 0. \quad (1.53)$$

Borrowing-limit multiplier. If $\tilde{\zeta}_s > 0$ for some s (the limit binds at equality), the household must devote the entire discounted stream of income to debt repayment,

$$\sum_{j \geq 0} R_{s,s+j} C_{s+j} = 0,$$

which implies $C_t = 0$ for $t \geq s$. By the Inada condition $\lim_{C \downarrow 0} U_c(C) = \infty$, this cannot be optimal. Hence $\tilde{\zeta}_t = 0$ for all t at the optimum.

Deriving the Euler Equation

Starting from (1.52) and dividing by β^{t-t_0} , with $\tilde{\zeta}_{t+1} = 0$:

$$-\frac{\lambda_t}{1+i_t} + \beta \lambda_{t+1} = 0 \Rightarrow \lambda_t = \beta(1+i_t)\lambda_{t+1}.$$

Substituting $\lambda_t = \frac{U_c(C_t)}{P_t}$ gives:

$$U_c(C_t) = \beta(1+i_t) \frac{P_t}{P_{t+1}} U_c(C_{t+1}).$$

Real form:

$$\underbrace{U_c(C_t)}_{\text{marginal utility of consumption at } t} = \underbrace{\beta(1+r_t)U_c(C_{t+1})}_{\text{marginal utility of saving to } t+1}, \quad 1+r_t \equiv (1+i_t) \frac{P_t}{P_{t+1}}.$$

$$U_c(C_t) = \beta(1+r_t) U_c(C_{t+1})$$

[Back to Euler Equation](#)

On the borrowing-limit multiplier

If the borrowing limit binds at equality at some s ,

$$-\frac{B_{s-1}}{P_s} = \frac{X_{s-1}}{P_s} + \sum_{j=0}^{\infty} R_{s,s+j} \left(Y - \frac{T_{s+j}}{P_{s+j}} \right),$$

then the RHS of the intertemporal budget from s onward is 0, so

$$\sum_{j=0}^{\infty} R_{s,s+j} C_{s+j} \leq 0 \Rightarrow C_t = 0 \quad \forall t \geq s,$$

which cannot be optimal under Inada ($\lim_{C \downarrow 0} U_c(C) = \infty$). Hence the borrowing limit does not bind at the optimum: $\tilde{\zeta}_t = 0$ for all t .

Proof: Expenditure Equation

We start from the budget constraints and market clearing:

$$\text{Household:} \quad \frac{B_t}{1+i_t} + \frac{X_t}{1+i_t} + P_t C_t + T_t = B_{t-1} + X_{t-1} + P_t Y, \quad (1.54)$$

$$\text{Central bank:} \quad X_{t-1}^C + \frac{B_t^C}{1+i_t} + T_t^C = \frac{X_t^C}{1+i_t} + B_{t-1}^C, \quad (1.55)$$

$$\text{Treasury:} \quad B_{t-1}^F = T_t + T_t^C + \frac{B_t^F}{1+i_t}, \quad (1.56)$$

$$\text{Market clearing:} \quad B_t^F = B_t^C + B_t, \quad X_t^C = X_t. \quad (1.57)$$

Eliminate T_t^C from (1.55) with (1.56):

$$X_{t-1}^C + \frac{B_t^C}{1+i_t} + B_{t-1}^F - T_t - \frac{B_t^F}{1+i_t} = \frac{X_t^C}{1+i_t} + B_{t-1}^C.$$

Use (1.57) to substitute $B_t^F = B_t^C + B_t$ and $X_t^C = X_t$, simplify to

$$\frac{B_t + X_t}{1+i_t} = B_{t-1} + X_{t-1} - T_t. \quad (1.58)$$

Insert (1.58) into the household budget (1.54):

$$P_t C_t = P_t Y \quad \Rightarrow \quad C_t = Y.$$

[Back to Euler Equation](#)